

University of Information Technology & Sciences (UITS)
Department of Computer Science & Engineering
BSc. in CSE — 4th Year, 1st Semester | Spring 2026
CSE 413: Simulation & Modeling Lab
Section: 7A, 7B

Total Marks: 10
Date of Submission: 29/06/2026

Target: CO3 & CO4, PO(c): Design/ Development of Solutions and PO(d): Investigation.

Course Outcome (CO)	Program Outcome (PO)	Knowledge Profile (K)	Complex Engineering Problem (P)
CO3: Construct a model for a given set of data and examine its validity.	PO(c): Design/ Development of Solutions	K2, K3, K5	P1, P2, P3
CO4: Analyze output data produced by a model and test the validity of the model.	PO(d): Investigation	K8	P1, P2, P3

Assignment Title: Designing a Simulation-Based Smart Traffic Analysis System for Urban Road Management

Background: SmartCity Traffic Solutions Ltd. (STSL) is a transportation technology company working to improve traffic management systems in major cities of Bangladesh.

The company collects traffic information from road sensors, CCTV systems, and GPS-based vehicle tracking applications. The collected data includes vehicle arrival rates, waiting times, and traffic flow patterns.

However, the existing traffic monitoring system faces several challenges:

- Traffic congestion patterns change randomly depending on time and location.
- Manual analysis of traffic data cannot predict future congestion conditions.
- Traffic simulation models often fail because they do not correctly represent real-world uncertainty.
- Different roads have different vehicle arrival distributions.
- Existing systems cannot estimate how traffic conditions will change after modifying traffic signals.
- Engineers need a reliable simulation system before implementing expensive road infrastructure changes.

To solve these problems, STSL wants to develop a simulation-based traffic analysis model that can generate traffic scenarios and evaluate possible solutions before real implementation.

The Engineering Challenge:

The company has assigned you as a Simulation Engineer to design a computational traffic simulation model.

Your objective is to create a model with python that can:

- Analyze collected traffic data.
- Identify traffic arrival patterns.
- Generate simulated traffic flow data.
- Compare simulated and real traffic behavior.
- Validate whether the simulation accurately represents real traffic conditions.

The challenge is complex because traffic systems contain uncertainty due to:

- Random vehicle arrival
- Different driving behaviors
- Peak-hour variations
- Unpredictable traffic flow

Multiple modeling approaches are possible:

- Random number-based simulation
- Normal distribution modeling
- Monte Carlo simulation
- Statistical hypothesis testing

You must investigate, design, implement, and justify your selected simulation approach.

Complicating Constraints:

- Traffic data that you chose should contain randomness and unexpected variations.
- The simulation should represent realistic vehicle arrival behavior.
- Incorrect distribution assumptions may produce inaccurate traffic predictions.
- The model must handle different traffic scenarios.
- Simulation results must be validated statistically.
- The developed model should support future smart-city applications.
- The solution should reduce unnecessary real-world testing costs.

Requirements:

Table 1: Functional Requirements

ID	Module	Description
FR-01	Traffic Data Analysis	Analyze vehicle arrival and waiting-time datasets using statistical measures.
FR-02	Traffic Simulation Module	Generate synthetic traffic data using random sampling techniques.
FR-03	Probability Modeling	Select suitable probability distributions for traffic behavior modeling.
FR-04	Monte Carlo Simulation	Simulate traffic scenarios using repeated random experiments.
FR-05	Model Validation	Apply statistical tests to verify simulation accuracy.
FR-06	Visualization Module	Plot original and simulated traffic distributions.
FR-07	Performance Analysis	Compare simulation results and analyze traffic behavior.
FR-08	Decision Support Module	Discuss how the simulation can help traffic planning decisions.

Non-Functional Requirements

- NFR-01 — Accuracy : The simulation model should generate traffic behavior close to real-world observations.
- NFR-02 — Reliability : The model should provide consistent output under repeated simulation runs.
- NFR-03 — Computational Efficiency :The system should efficiently process large numbers of simulated vehicles.

- NFR-04 — Validation Capability : Simulation results must be verified using statistical techniques.
- NFR-05 — Scalability : The model should support future expansion for smart-city traffic systems.
- NFR-06 — Engineering Interpretation : The simulation output should provide meaningful traffic insights.

Assessment Rubrics (Total: 10 Marks)

#	Assessment Criterion	Indicators of Achievement	Marks
1	Simulation Model Design	2 = Proper traffic simulation approach selected with clear workflow and justification. 1.5 = Minor gaps. 1 = Incomplete design. 0.5 = Incorrect approach. 0 = Not submitted.	0–2
2	Data Analysis & Distribution Modeling	2 = Correct statistical analysis and probability modeling with proper interpretation. 1.5 = Minor mistakes. 1 = Limited analysis. 0.5 = Major gaps. 0 = Not submitted.	0–2
3	Validation & Statistical Testing	2 = Correct Chi-square testing, p-value analysis, and conclusion. 1–1.5 = Partial validation. 0.5 = Weak testing. 0 = Not submitted.	0–2
4	Engineering Investigation	2 = Strong discussion of assumptions, limitations, and improvements. 1 = Basic analysis. 0.5 = Weak justification. 0 = Not submitted.	0–2
5	Report Quality & Visualization	2 = Clear graphs, explanation, and technical presentation. 1 = Minor issues. 0.5 = Poor reporting. 0 = Not submitted.	0–2

Marking Note:

Evaluation will consider both technical correctness and engineering reasoning. Students must justify their simulation choices and explain how the model represents the real-world system. **Plagiarism or identical submissions will receive zero marks. Your individual model should not resemble others' models.**